

MATH 4360 HOMEWORK ASSIGNMENT 3

DUE ON THURSDAY 14 SEPTEMBER 2023

Please turn in your homework in class or to Weeks Hall 347 no later than 5 pm.

- (1) (a) Show that $x^3 + x + 1$ is irreducible over $\mathbb{Z}_2$.
(b) Calculate the addition and multiplication tables for the field

$$F = \mathbb{Z}_2[x]/\langle x^3 + x + 1 \rangle.$$

- (c) Find a generator for the multiplicative group F^* .

- (2) Factor $x^4 - 1$ as a product of irreducible polynomials in $\mathbb{Q}[x]$.

- (3) Find all irreducible polynomials of degree at most 3 over $\mathbb{Z}_2$.

- (4) Explain why there are exactly 171 irreducible monic quadratic polynomials over $\mathbb{Z}_{19}$.

- (5) Decide if the polynomial $x^4 - 5x^3 + 2x + 3$ is irreducible over $\mathbb{Q}$.